Keysight PROPSIM

S9165A Test System Setup Guide

Application note

KEYSIGHT TECHNOLOGIES

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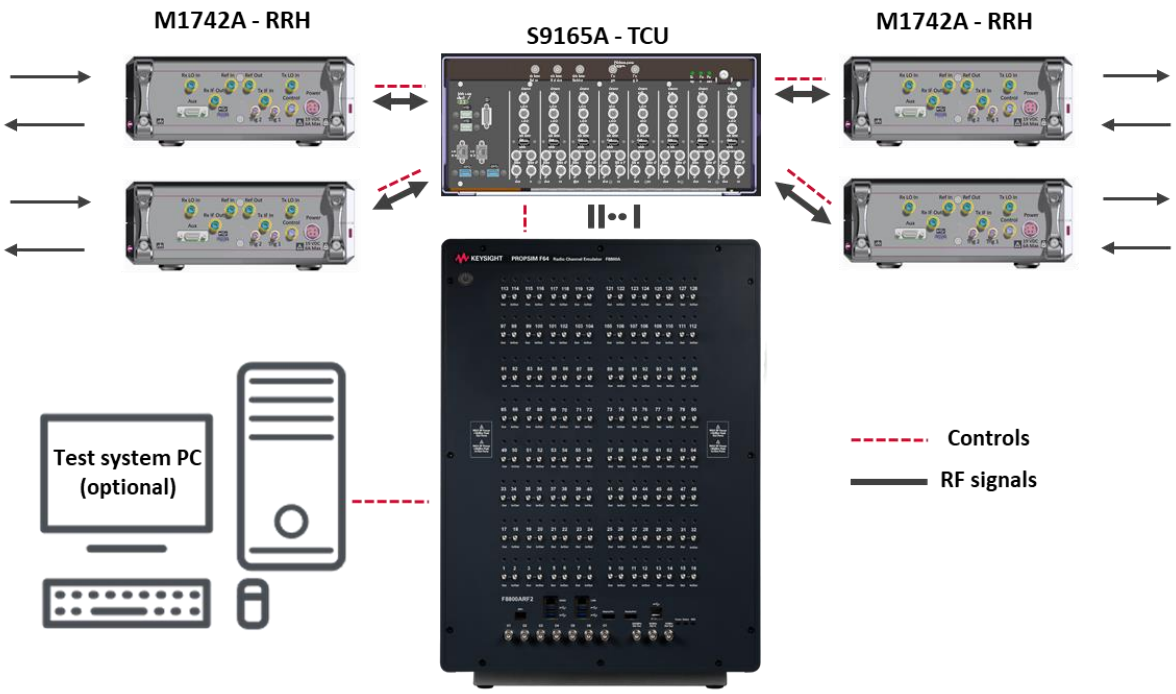
1 INTRODUCTION

The Keysight S9165A Test System consists of the following components:

- PROPSIM F64/FS16 (1 pc)
- S9165A (1 pc)
- M1742A RRH (1-8 pcs)



1.1 S9165A

S9165A unit controls the remote radio head and communicates with PROPSIM. The unit front panel has connections to both PROPSIM and RRH units. Place the S9165A unit close to PROPSIM. Each remote radio head connects to the unit with a cable harness that is 4 meters long, so ensure that the unit is close enough to PROPSIM and remote radio heads.

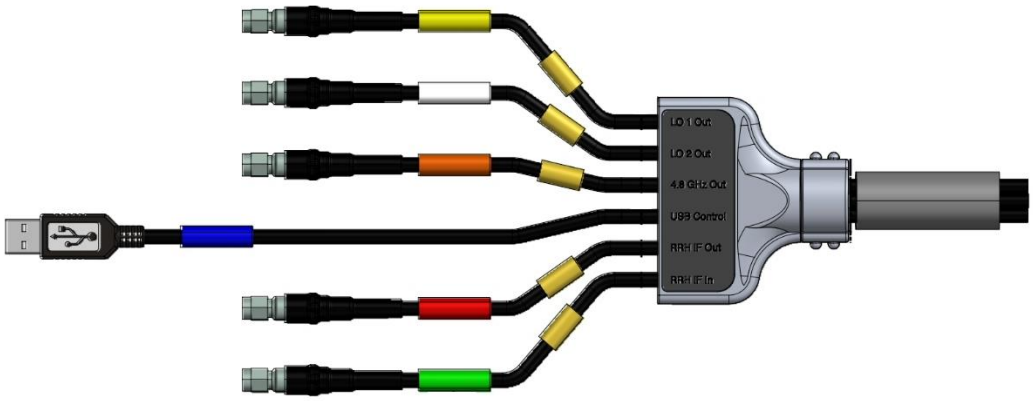
1.1.1 Cable connections

S9165A with the front panel connections can be seen in the figure below.

Number	Name	Connection
1 & 2	2x USB 2.0	USB connectors for peripheral connection
3	DisplayPort	DisplayPort connector for external monitor
4 & 5	2x LAN	Connectors for ethernet connection
6 & 7	2x USB 3.0 SuperSpeed	USB connectors
8	10 MHz Ref In	10 MHz reference frequency connector (from PROPSIM)
9	10 MHz Ref Out	10 MHz reference frequency output connector
10	100 MHz Ref Out	100 MHz reference frequency output connector
11 & 12	Trig 1 & Trig 2	Not used
13	LO 1 Out	First LO signal output to M1742A RRH
14	LO 2 Out	Second LO signal output to M1742A RRH
15	4.8 GHz Out	4.8 GHz reference frequency output to M1742A
16	USB Control	USB control signal connector to M1742A
17	RRH IF Out	IF output connector (IF signal to M1742A)

Number	Name	Connection
18	IF In	IF input connector (IF signal to PROPSIM)
19	RRH IF In	IF input connector (IF signal from M1742A)
20	IF Out	IF output connector (IF signal from PROPSIM)

The remote radio heads will be connected to the corresponding connectors with the cable harness that has the according labels. The end of the cable that will be connected to the S9165A can be seen in the figure below.



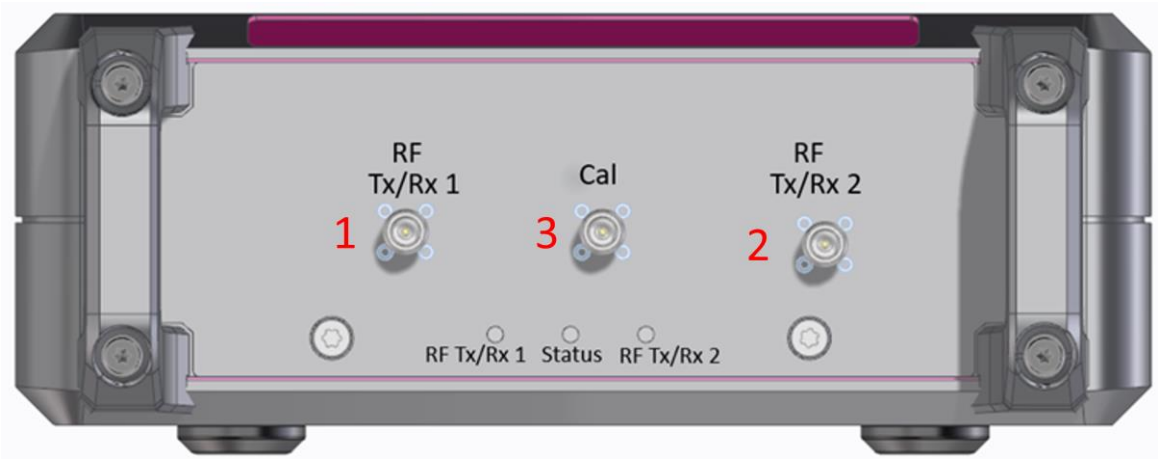
Connect each head in the order that is intended (first head connected to first channel) according to the labels in the cable.

1.2 M1742A RRHs

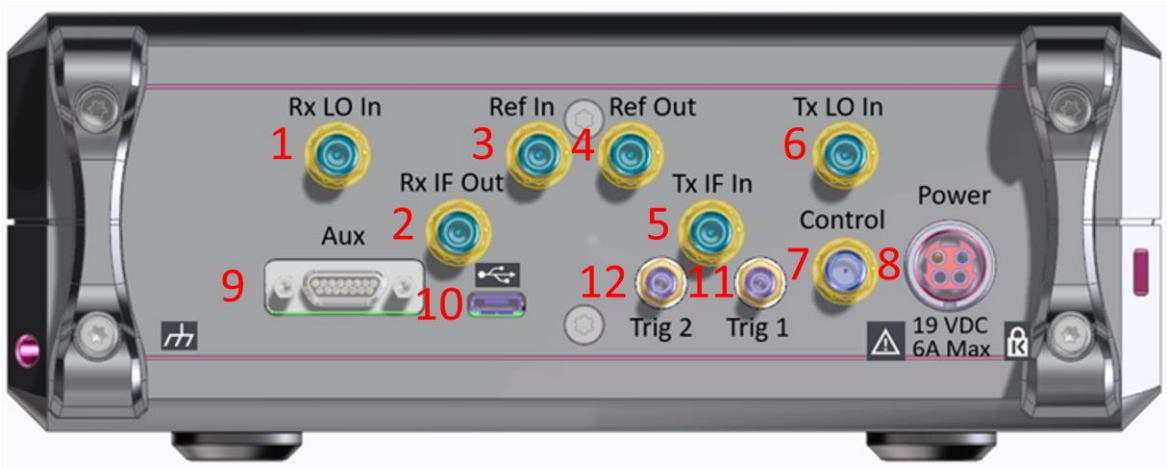
For higher frequency operation M1742A remote radio heads (RRHs) are provided. Each head has two mmWave connections and IF connectors as well as LO inputs. Using the provided cable harness and IF cables, the heads will connect to S9165A front panel and PROPSIM.

1.2.1 Cable connections

The M1742A connectors can be seen in the figures below. Following the figures are tables explaining the usage of the connector.

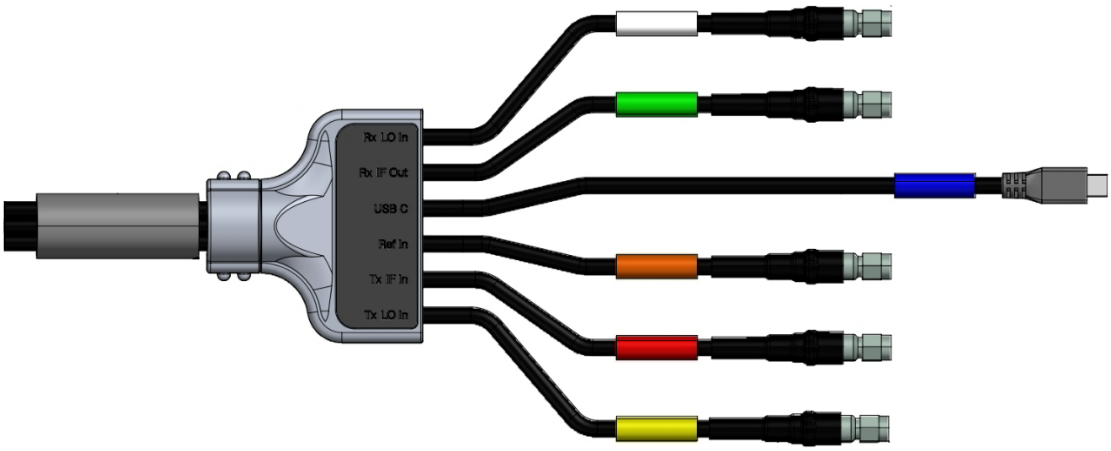


Number	Name	Connection
1	RF Tx/Rx 1 (2.4 mm connector)	A port which can be configured to transmit signal to probe antenna
2	RF Tx/Rx 2 (2.4 mm connector)	A port which can be configured to receive signal from probe antenna
3	Cal	Not used



Number	Name	Connection
1	Rx LO In (SMA connector)	Downconversion LO signal from S9165A
2	RX IF Out (SMA connector)	Downconverted Rx IF signal to S9165A
3	Ref In (SMA connector)	Reference frequency signal from S9165A
4	Ref Out (SMA connector)	Not used
5	Tx IF In (SMA connector)	IF signal to be upconverted from S9165A
6	Tx LO In (SMA connector)	Upconversion LO signal from S9165A
7	Control	Not used
8	Power	19 V/6A DC power from proprietary power supply
9	Aux	Not used
10	USB	Micro-USB Type-AB control connector from S9165A
11 & 12	Trig 1 & Trig 2	Not used

Connect the wiring harnesses as labelled in the connectors. The end of the cable that will be connected to the remote radio heads can be seen in the figure below.



Connect the corresponding channel to the correct RRH by following the labels in the cable.

1.3 Important to note

- The ethernet cable to PROPSIM must be connected to the provided USB3.0 – gigabit ethernet adapter. The ethernet adapter must be connected to a free USB3.0 port on the PROPSIM front panel. Do not connect the cable directly to the ethernet connector on the PROPSIM front panel. The integrated PROPSIM ethernet connector is left for free customer use.
- The M1742A remote radio heads ship with the proprietary power supply which must be used for powering the RRH units.
- The 10MHz reference BNC cable must be connected to the “10MHz Out” connector on the PROPSIM front panel and the “10MHz In” connector on the S9165A front panel.
- When tightening the SMA connectors, use the provided torque wrench to ensure a reliable RF path.
- Ensure that the S9165A is placed so, that there is enough space around the device for proper ventilation.
- Do not place the M1742A RRH units close to each other side-by-side to ensure sufficient cooling.
- Do not put mechanical stress and/or strain on the connectors.
- Do not hang any device by the connector.
- To ease the cabling, hold the connector the right way up so that the connectors line up in the correct order to match the S9165A and M1742A RF ports.

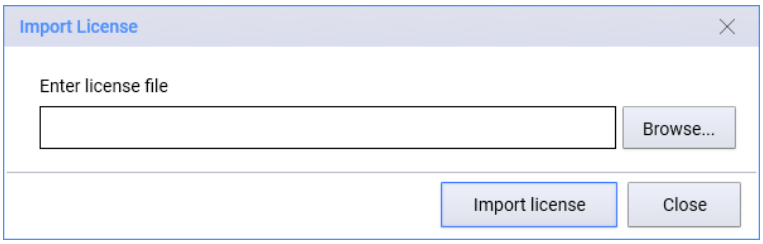
2 PROPSIM CONFIGURATION

This chapter explains what needs to be set up in the PROPSIM to enable it to control the S9165A system. The PROPSIM platform version must be Release 9 or newer to be able to control the S9165A and the M1742A RRH system.

2.1 Importing License

The license file activates the extended frequencies in PROPSIM.

1. In the PROPSIM user interface, open the **Import License** dialog by selecting **Configuration > Import license** in the navigation menu.



2. Browse and select the license file (file that has the word "lservrc" in it), and click the **Import license** button. The file containing license changes is compared to the current license files and a list of additions or modifications is shown in the **License File Comparison Result** dialog.
3. Accept the changes by clicking **OK** in the **License Files Comparison Result** dialog.

The original license files are first backed up in their current folder and the changes are written to the original license files. The backup files are named as "lservrc.backup_YYMMDD_HHMMSS", where YYMMDD describes the date of the backup and HHMMSS the time of the backup. Restart is required after license import.

The license configuration in use can be checked from the PROPSIM GUI side panel. Navigate to Help->Technical Support->My software support and ensure that the RRH license shows up. Below is an example row where a license unlocking the use of 4 RRH units is applied.

2.2 Ethernet configuration

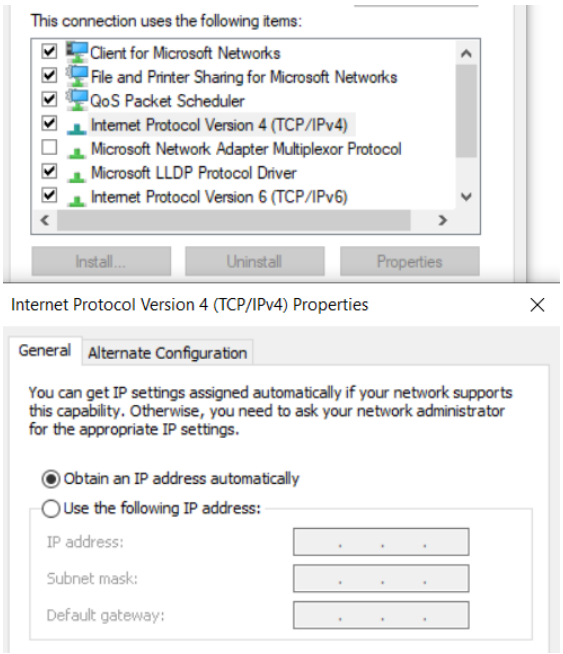
The provided USB3.0 – Gigabit ethernet adapter must be configured manually. In PROPSIM's Windows control panel (Control Panel->Network and Internet->Network and Sharing Center->Change adapter settings and select the USB adapter), change the USB3.0 – Gigabit ethernet adapter to use such parameters that the S9165A can be reached and the provided DHCP service associated with the external unit driver can be used. This means that only the gigabit ethernet adapter must be configured and the S9165A IP address can be set automatically. The following configuration must be used for the ethernet adapter.

PROPSIM USB3.0 – Gigabit ethernet adapter

- Ipv4 address: 10.168.0.1
- Subnet mask: 255.0.0.0

S9165A

- Set the Ipv4 address to be obtained automatically, see the figure below. This will finally be obtained from the DHCP server, and set to 10.168.0.90. Alternatively, the Ipv4 address can be set manually using the properties below.
  - Ipv4 address: 10.168.0.90
  - Subnet mask: 255.255.255.0



2.3 External unit driver

You can download the external unit driver software from the Keysight Software Manager (http://www.keysight.com/my). This requires a myKeysight account. The installer also contains a configuration file with a default OTA testing setup with eight M1740A RRH units and a E7770A CIU.

To install the external unit driver software, run “Propsim_External_Units_Driver_Installer.exe” as administrator. The driver also installs the configuration file “externalUnit.config” into the folder C:\PROPSIM\ExternalUnit\.

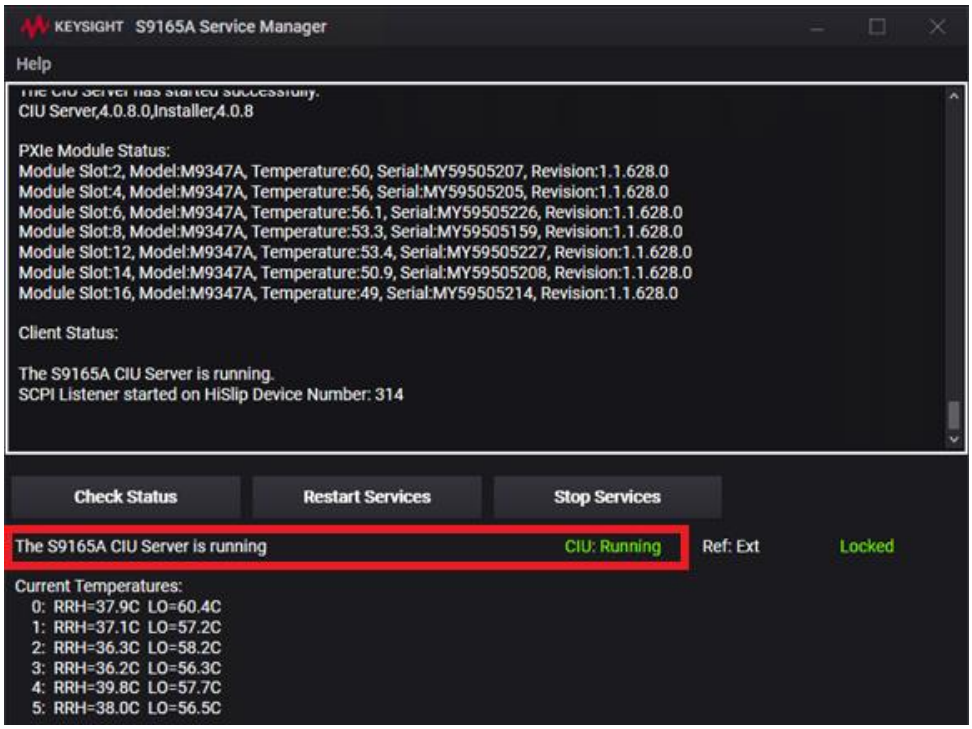
The configuration file must match the used setup (number of RRHs, RRH ports in CIU/PROPSIM/S9165A and possible chamber rotators). Refer to the PROPSIM User Reference for information on creating the external unit configuration file.

NOTE: If you have a predefined configuration file provided by Keysight, copy the file (ExternalUnit.config) to C:\PROPSIM\ExternalUnit and reboot PROPSIM.

2.4 Starting up devices

After the licenses, external unit driver, and configuration file are installed, start up the devices exactly as instructed in the following steps:

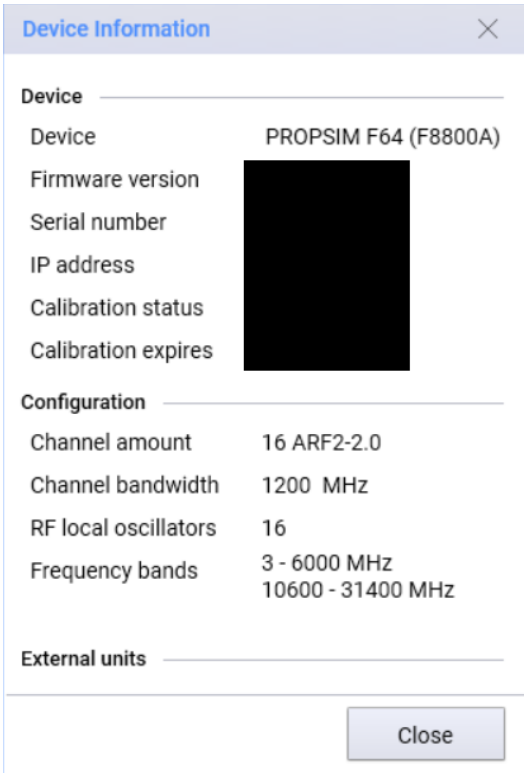
1. Shut down the PROPSIM by selecting **System > Restart / Shutdown > Shutdown** in the navigation bar. Click **OK** in the confirmation dialog.
2. Check the cable connections to make sure the connections are OK.
3. Turn on the S9165A unit. Please wait up to 10 minutes for the unit to startup. The status of the S9165A bootup can be checked by inspecting the S9165A Service Manager GUI. After the following information is shown in the GUI, PROPSIM can be turned on.



4. Turn the PROPSIM back on

Note: *The S9165A must be powered on before PROPSIM in every start-up, not only in the first start-up.*

When everything is working as expected, the PROPSIM GUI opens without errors. The **Device Information** dialog shows the extended frequencies (**Configuration > Device Information**). If the extended frequencies are used in an emulation, PROPSIM will automatically use the RRH units.



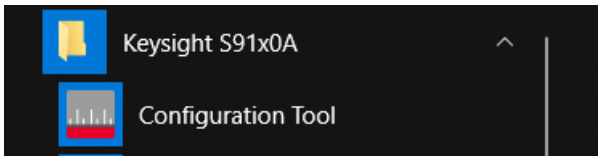
At start-up, some ports in the emulator front panel are reserved for the extended frequencies (RRH units).

3 S9165A CONFIGURATION

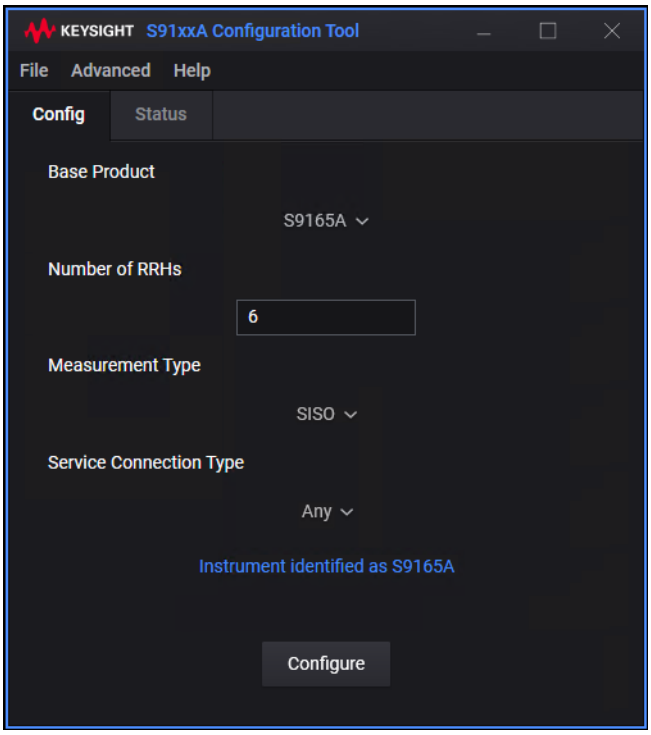
Keysight S9165A control unit has its own control software (S91XXA software) that needs to be configured when setting up the device. After initial configuration the system has to be rebooted for the changes to take effect. S9165A integrates a Windows PC which can be controlled either remotely or by connecting the peripherals to the unit front panel.

3.1 Configuration

To configure the S9165A system, run the S91XXA Configuration Tool from Start-menu (Start->Keysight S91x0A->Configuration Tool).



To configure the system correctly, first select the type of base product in use. For S9165A configuration either leave the setting to 'Auto' or set it to be 'S9165A'. After this, adjust the number of RRH units to correspond to the number of M1742A remote radio heads connected to the system. Measurement Type-field can be left to 'SISO', but the Service Connection Type should be set to 'Any'. Below are the settings for 6 RRH configuration.



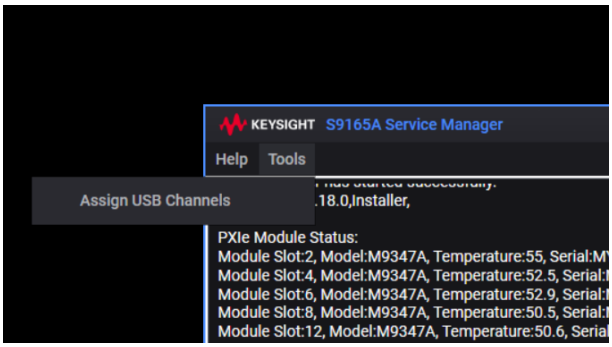
After configuration, the Configure-button can be clicked. If the 'Configuration successful!'-message is shown, the system will boot up correctly during the next boot.

3.2 RRH mapping

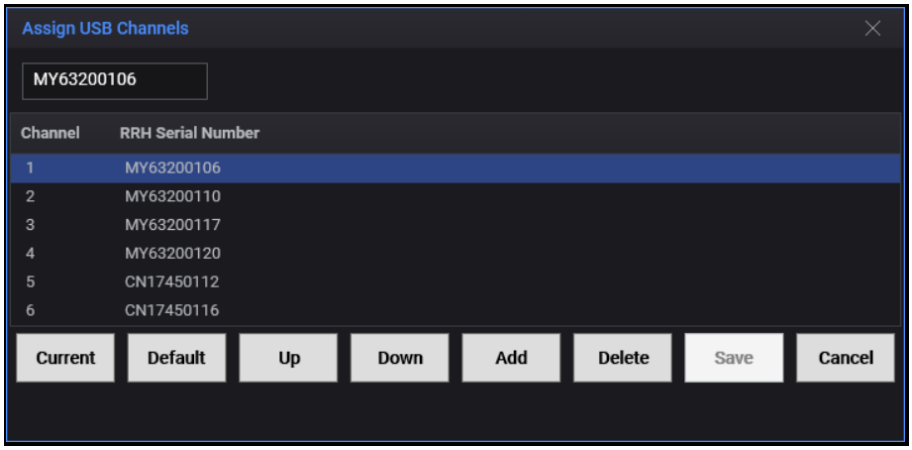
Due to the RRH units having a USB connection to S9165A unit the RRH order compared to the channel numbers must be defined by the user. The order should always match in the way that they are connected to the S9165A channels. In practice, the head connected to the first channel of the S9165A should be mapped as the first RRH, the head connected to the second channel should be mapped as the second RRH etc. Following subchapters describe two ways to define the order.

3.2.1 Assigning USB channels from Service Manager UI (Service Manager version >= 4.0.18)

To ensure that the RRH units remain in the correct order over S9165A boots, a configuration dialog can be used in the newer version of the Service Manager to define the order. The dialog can be opened from the Service Manager UI from Tools -> Assign USB channels.



Below is a screenshot of an example system that has 6 RRH units connected to it. The serial numbers of each RRH unit should show in the dialog, and if not, check the connection and power status of the RRH not showing up.



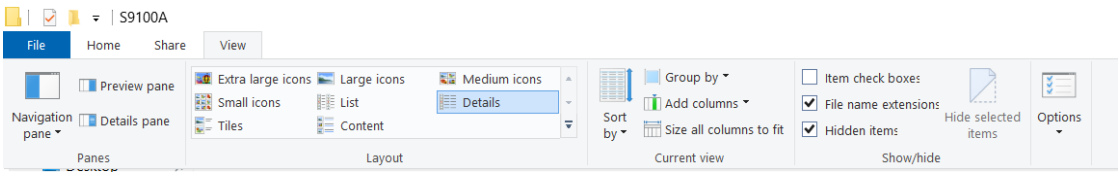
Using the buttons the order can be defined;

- **Current:** Load the existing RRH serial number mapping if it exists
- **Default:** Load the RRH serial number mapping from the scanned RRH serial numbers found in the system bootup.
- **Up:** Move the selected entry up one position in the list.
- **Down:** Move the selected entry down one position in the list.
- **Add:** Add a new entry to the bottom of the list. Maximum number of entries is 8. The serial number of the new entry is set to "<none>" by default, and should be edited to match the desired RRH unit serial number.
- **Delete:** Remove the selected entry from the list.
- **Save:** Save the current serial number list to be used in the next bootup.
- **Cancel:** Close the dialog.

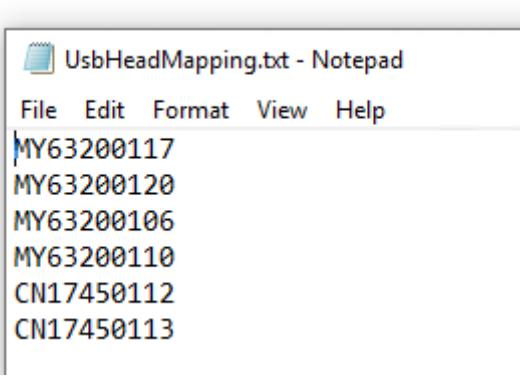
After modifying the order, the Service Manager has to be restarted in order to apply the effect of the modifications.

3.2.2 Modifying the .txt-file (Service Manager version < 4.0.18)

To ensure correct order of the RRH units, they should be manually configured. Manual configuration consists of modifying a text file in S9165A file system. The file to edit is C:\ProgramData\Keysight\S9100A\UsbHeadMapping.txt. Note that ProgramData-directory is a hidden folder. Hidden directories will not show up by default in Windows Explorer, but this can be enabled from File Explorer toolbar by checking View->Hidden Items as shown in the figure below.



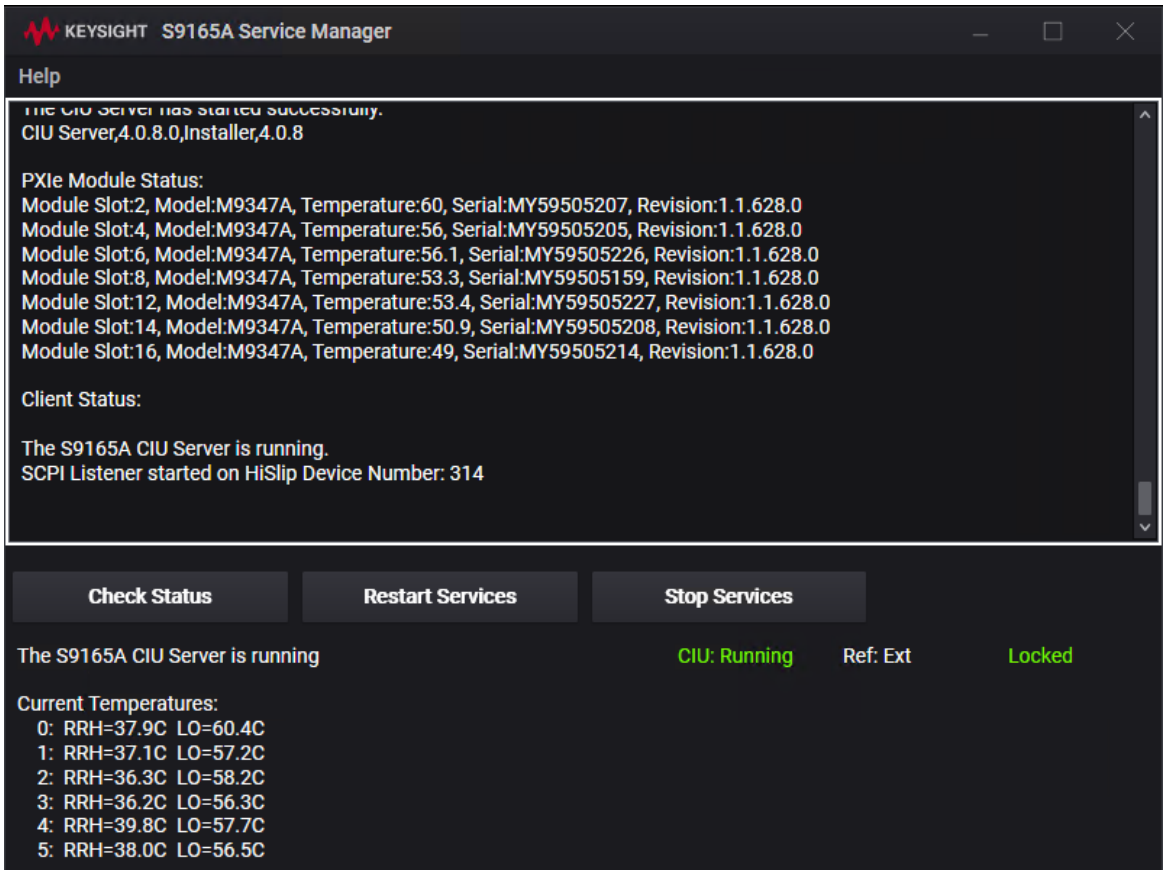
To set the order of the radio heads, open the .txt-file for edit and enter the RRH serial numbers in the order that they are connected to the S9165A channels. Each serial number should be written to its own line. For example, in the example below two heads are connected to the S9165A system, the one with the serial number MY63200117 wired to the first channel and the one with the serial number MY63200120 wired to the second channel etc. The serial number of the M1742A RRH can be found under each RRH unit.



After configuring the serial numbers, the .txt-file can be saved and the S9165A system rebooted.

3.3 Overview

After reboot, the system will start the S9165A Service Manager-application starting the control unit software. The boot can take several tens of minutes, after which the application interface should look similar to the screenshot below. Note that the example system consists of six RRH units.



After successful start the bottom text should state 'CIU: Running'. The reference information can show also 'Int', but this should change after successful PROPSIM startup. After the initial configuration explained in this chapter the peripherals can be disconnected and the S9165A system will be controlled by PROPSIM. If the configuration changes (heads are re-cabled, head count changes etc.) the configuration procedure will have to be redone.

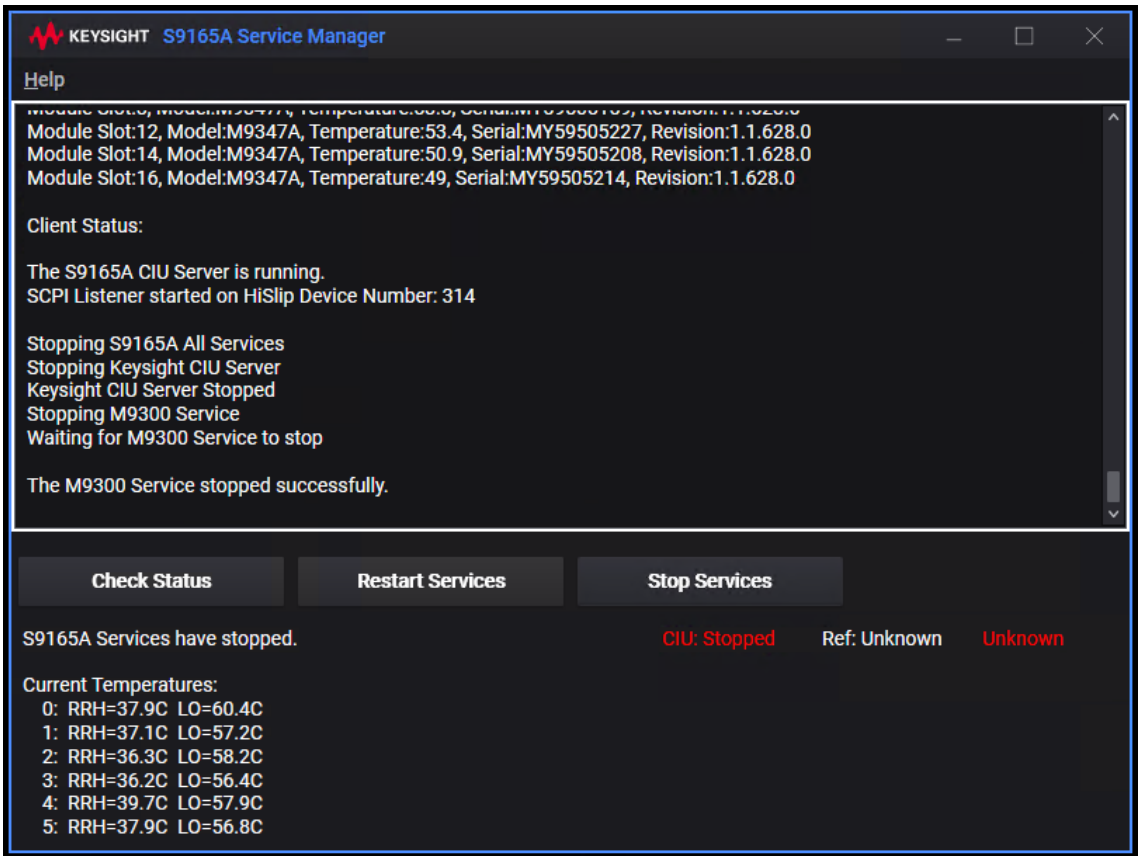
4 RECONFIGURING

If the setup changes, refer to the following chapters in order to reconfigure the system correctly.

4.1 Adding/removing RRH

As a first step, update the external unit configuration in PROPSIM to match the setup that will be used. After this step, PROPSIM can be turned off.

As a second step, the existing S9165A software must be shut down to start the reconfiguration. From the S9165A Software Manager, click the button “Stop Services” to shut down the software. When the text “S9165A Services have stopped.” appears on the window, close the UI window. See the figure below how the stopped-state of the software should look like.



To reconfigure the S9165A software, first run the S91xxA Configuration Tool by following the instructions described in chapter 3.1. After this, the Configuration Tool can be closed and the serial numbers updated by following chapter 3.2. Before proceeding with rebooting of the devices, remove the file RegisteredClients.xml from the directory C:\ProgramData\Keysight\S9100A. After this the reconfiguration is done and the S9165A system can be rebooted. After a successful reboot of the S9165A system, PROPSIM can be booted up again.

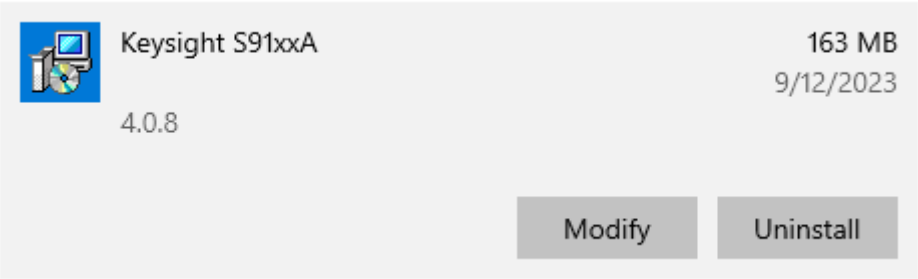
4.2 RRH reordering

If the number of heads does not change but the heads are switched in order, follow the instructions described in chapter 3.2. NOTE: The order cannot be changed by only remapping the serial numbers. The order of the heads given with the .txt-file must match the order that the heads are cabled to the S9165A channels. Also, make sure that the indexing matches the order in the PROPSIM external unit configuration (head with the correct index is cabled to the correct S9165A channel).

4.3 S9165A software update

In order to update the S9165A, the existing software must be shut down and uninstalled before running the updater. Also, before starting the update, PROPSIM must be turned off.

To update the software of the S9165A, shut down the S9165A Service Manager by clicking “Stop Services”-button. After the window states “S9165A Services have stopped.”, the UI window can be closed. Navigate to “Apps & features” in Windows settings (Settings->Apps) and uninstall the “Keysight S91xxA”.



After successful uninstallation, update the software by running the provided/downloaded executable. After update, reboot the S9165A, run the configuration procedure again (described in chapter 3).

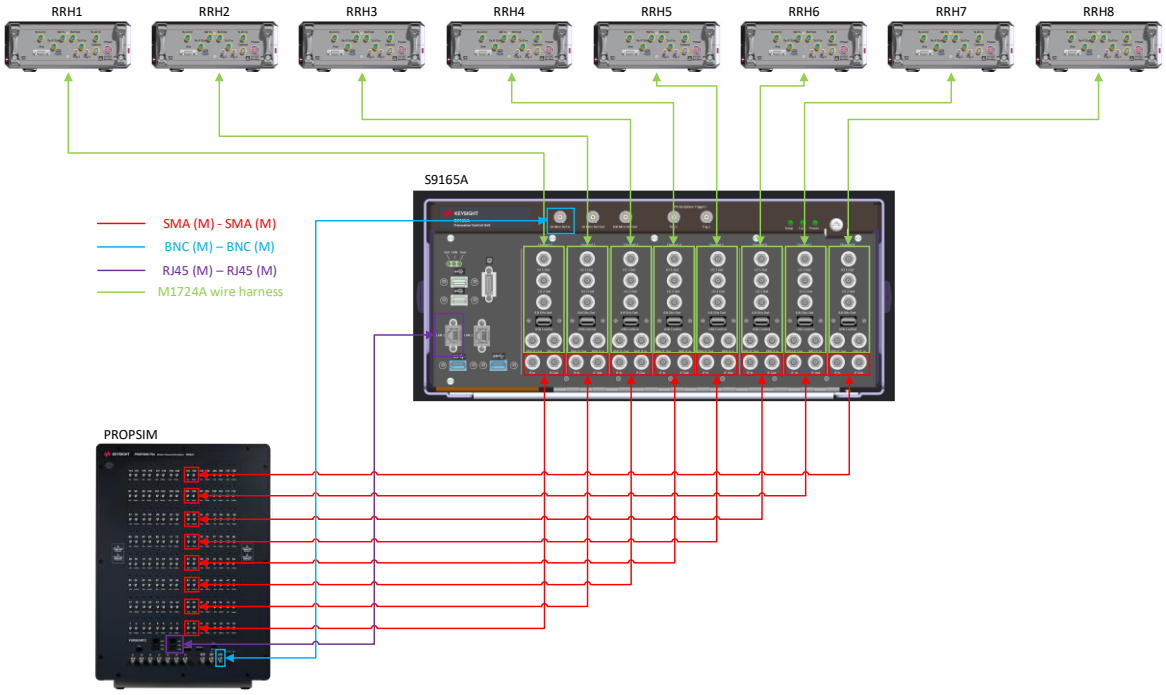
After update and reconfiguration of the S9165A, PROPSIM can be turned on again.

5 CONFIGURATION EXAMPLE

In this chapter an example configuration of a S9165A, 8 M1742A RRH units and 8 CU PROPSIM F8800A is configured. The setup is configured to utilize PROPSIM ports that enable the usage of wide bandwidths up to 1200 MHz.

5.1 Cabling

After unboxing the equipment, the setup can be started with cabling. All the devices must have power cords connected appropriately. RRH units are provided with proprietary power supplies. After powering the units, see the illustration below on how to cable each RRH and the S9165A and PROPSIM.



Connection to the RRH units (green) should be made with the provided wire harnesses, refer to 1.1.1 and 1.2.1 for details on the connectors. IF signals (red) should be connected with the provided SMA-SMA-cables, S9165A "IF In" connected to PROPSIM outputs (RF connectors with odd numbering) and S9165A "IF Out" connected to PROPSIM inputs (RF connectors with even numbering). RRH1 should be connected to first PROPSIM channel unit, RRH2 to second etc. 10 MHz reference frequency (blue) should be connected from PROPSIM "10 MHz Out" to S9165A "10 MHz Ref In" by using a BNC-BNC-cable. LAN connection (purple) should be made with the provided USB network card from PROPSIM to S9165A.

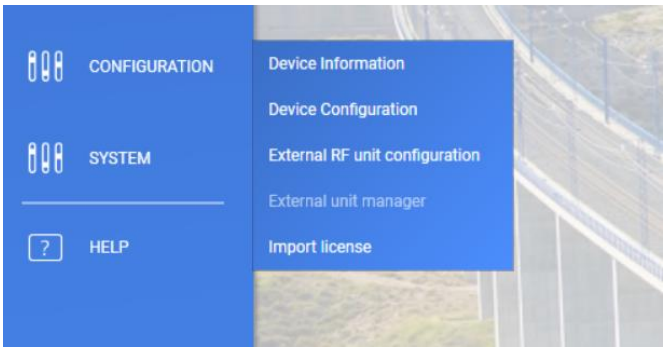
5.2 S9165A configuration

After cabling the devices, the S9165A can be powered on by pressing the power button. To access the embedded Windows PC, connect peripherals (keyboard, mouse) and a monitor to the unit. Follow the instructions provided in 3.1 and 3.2 to configure the system. Before proceeding to PROPSIM configuration make sure that the S9165A has started correctly and all 8 RRH units are shown in the Service Manager UI.

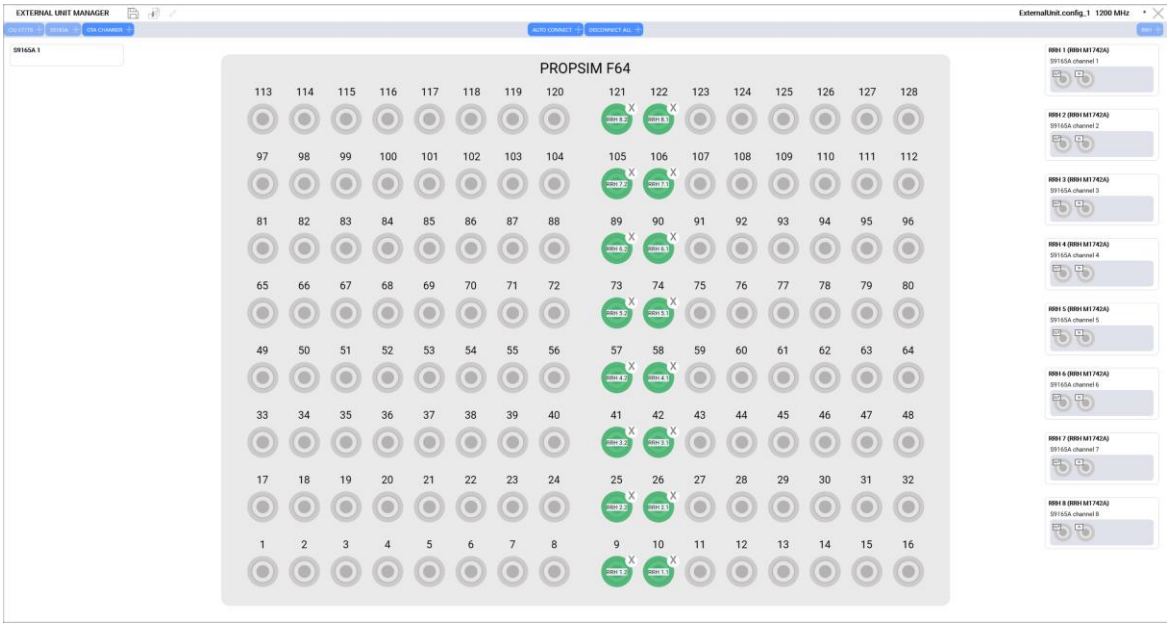
5.3 PROPSIM configuration

After the cabling and S9165A configuration PROPSIM can be configured to reserve the correct ports to RRH usage and unlocking the higher frequencies. Before the external unit configuration, make sure that the LAN interface is configured correctly by following the instructions in 2.2. The connection can be tested by for example trying to remotely connect to the S9165A IP address using Windows Remote Desktop Connection.

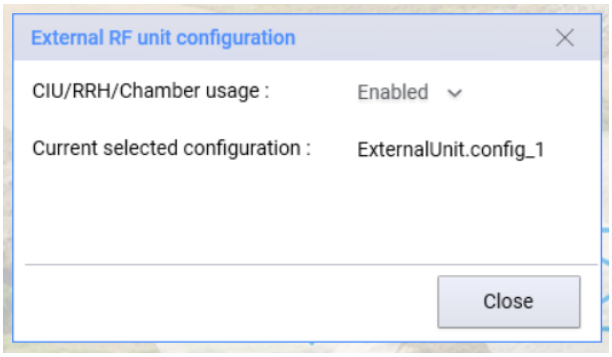
When the LAN interface has been configured, the external unit configuration can be made. Navigate by clicking from PROPSIM left-hand side panel to Configuration->External unit manager.



Follow the instructions provided in PROPSIM User Reference chapter 17 to create the setup. Final setup should look like the one below in the picture, which will match the cabling.

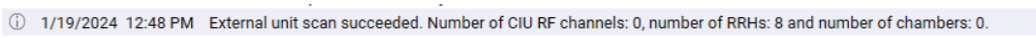


Take the configuration into use by clicking the button on the right hand side of the save-button in the upper left hand corner of the external unit manager UI. Make sure that the correct configuration is in use by navigating from PROPSIM left hand side panel to Configuration -> External RF unit configuration. It should look like the dialog in the screenshot below.



Please note that the configuration that was used here was named "ExternalUnit.config_1", make sure that the configuration name matches the one created in the previous phase in External unit manager. Before proceeding with PROPSIM restart (System->Restart (power cycle)), make sure that the S9165A is powered on and the Service Manager has started correctly.

After successful reboot of PROPSIM, the PROPSIM UI log should show the following text;

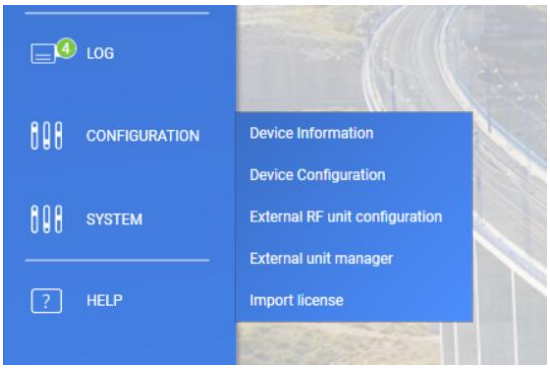


The correct frequency ranges should be available in emulation creation and should also show up in Configuration->Device Information.

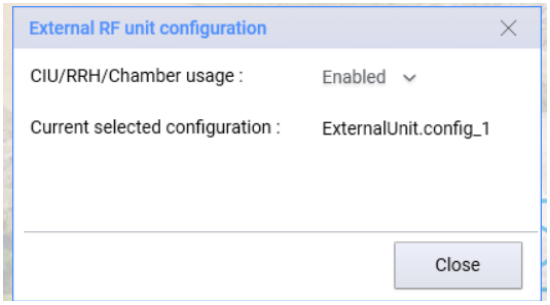
6 TROUBLESHOOTING

If the system does not control the RRH units correctly or there are problems while starting up the system, check the following:

- Firewall rules (Keysight CIU server must be allowed)
  - In S9165A, navigate to Control Panel->System and Security->Windows Defender Firewall->Allowed apps
  - Check that the CIU Server application is allowed in the current network configuration
  -

- Check the IP address settings
  - Check that the IP address settings in both devices (PROPSIM and S9165A) matches the ones instructed in chapter 2.2.
    - IP address can be checked by for example running *ipconfig* in Command Prompt
  - Check that the external unit configuration has the correct IP address (10.168.0.90)
- Check the Service Manager status
  - Service Manager GUI can be inspected to check the status
  - Click the "Check Status"-button and see if the CIU Server is running and there is no other errors in the log window
- Check the wiring harness
  - The wiring harness must be cabled so that the head with the correct order (index) is cabled to the corresponding S9165A channel and the connections must match the labels
- Restart Service Manager
  - If something is wrong in the Service Manager, the software can be reset by clicking the "Restart Services"-button from the S9165A Service Manager GUI
  - NOTE: Restarting the S9165A software requires PROPSIM reboot
- If any of the RRH units show a temperature of 0.0 ° C, check that the RRH is powered and connected via USB
  - M1742A has no power button but the power supply must be powered on, the RRH fan generates some noise and in the RRH circuitry there should be LEDs that are lit up
- Remove Clients (Stop Services -> Close window -> remove RegisteredClients.xml -> Reboot)
  - From the S9165A Software Manager, click the button "Stop Services" to shut down the software. When the log text "S9165A Services have stopped." appears on the log window, click the red cross to close the UI window
  - Remove the file RegisteredClients.xml from the directory C:\ProgramData\Keysight\S9100A
  - Reboot S9165A and after a successful boot, reboot PROPSIM
- Check that the external RF units are enabled in PROPSIM
  - 1. Open the External RF unit configuration dialog (PROPSIM sidebar -> Configuration -> External RF unit configuration)
    - 

  - 2. Check that the units are enabled and the correct configuration is selected

- 
External RF unit configuration
CIU/RRH/Chamber usage :
Enabled
Current selected configuration :
ExternalUnit.config_1
Close